

# Constructing an Agent-Based Model

From Organized Complexity to Prospective Claims and Actionability

## Purpose

The central methodological problem is not simply how to program an ABM. It is:

**How do we decide what ABM to construct?**

Once the model exists, a second problem follows:

**What can we defensibly claim from it?**

The guide therefore moves from **ex-ante construction** to **ex-post accountability**:

**EX ANTE: CONSTRUCT → SIMULATE → EX POST: DOCUMENT & ASSESS**

### Source boundaries and course terminology

The six core readings support the theoretical foundations; citations distinguish their contributions from this course's synthesis. "Bottom-up + along the way," the Four As, the parsimonious I-ABM heuristic and the distinction between  $A_{3m}$  and  $A_{3s}$ , auditability/actionability, and the broad use of "meta-analysis" are course conventions. They are not attributed as a combined framework to the readings.

Ex ante and ex post distinguish methodological questions. Documentation can begin during construction, and later assessment can prompt revision.

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# 1 Foundations: Thinking About ABM Construction

Social systems present problems of **organized complexity**. ABM provides a way of constructing explicit generative models of such systems.

Two heuristics organize the construction process:

$$\boxed{\text{BOTTOM-UP}} + \boxed{\text{ALONG THE WAY}}$$

These are **not opposites**.

- **Bottom-up** is the generative backbone.
- **Along the way** identifies additional questions that arise as the model is progressively constructed.

## 1.1 Weaver — Organized Complexity

Start with the **phenomenon**, not with the computational agent. Weaver distinguishes simplicity, disorganized complexity, and organized complexity [1].

The problem is not merely that many things exist. They are organized and interdependent.

$$\boxed{\text{Social phenomenon} \rightarrow \text{organized complexity}}$$

**Question:**

What organization and interactions matter for the phenomenon?

## 1.2 Simon — Architecture of Complexity

Simon analyzes hierarchy and near-decomposability, with consequential differences between interactions within and between subsystems [2]. Complexity has organization:

- subsystems,
- hierarchy,
- differentiated levels,
- interdependencies,
- persistent structures.

$$\boxed{\text{Complexity} \rightarrow \text{Organization} \rightarrow \text{Structured interaction}}$$

Organization can generate persistence without implying complete predictability.

## 1.3 Bottom-Up: The Generative Backbone

### 1.3.1 Epstein — Generate

Epstein asks how a micro-specification can generate a macro-regularity [3, pp. 41–43]. Generation provides a candidate explanation; it does not uniquely identify the real mechanism. The generative question is:

**How could local interactions among heterogeneous autonomous agents generate the macro-pattern of interest?**

Micro-specification  $\rightarrow$  Interaction  $\rightarrow$  Macro-pattern

The modeller constructs the model, not its emergent outcome:

CONSTRUCT THE MODEL  $\neq$  CONSTRUCT THE OUTCOME

Therefore:

**Construct the conditions of possibility, not the emergent result.**

### 1.3.2 Cioffi-Revilla — Construct Progressively

Cioffi-Revilla motivates theoretically guided model sequences with inspiration from Lakatos [4]. The course develops this into explicit ontological re-specification, including removal and reclassification. Begin with a deliberately restricted model and progressively re-specify it:

$O_0 \rightarrow O_1 \rightarrow O_2 \rightarrow \dots$

Progressive construction is **not**:

$$O_{n+1} = O_n + \text{more detail}$$

Instead:

$O_{n+1} = \text{re-specify}(O_n)$

using operations such as:

ADD REMOVE DIFFERENTIATE RECLASSIFY RECOMBINE

Thus:

**Progressive construction = progressive ontological re-specification**

## 1.4 Along the Way: Situating the Model

As the model develops, additional questions arise.

### 1.4.1 Knoeri / Giddens — Situate Agency

**Agency is situated.**

Knoeri et al. operationalize context-specific agents, interactions, decisions, and behavior, drawing on Giddens and other foundations [5]. Agents act in relation to:

- other actors,
- resources,
- environments,
- technologies,
- social relations,
- institutional conditions.

The modeling question is therefore:

## Whose endogenous agency must be represented for this question?

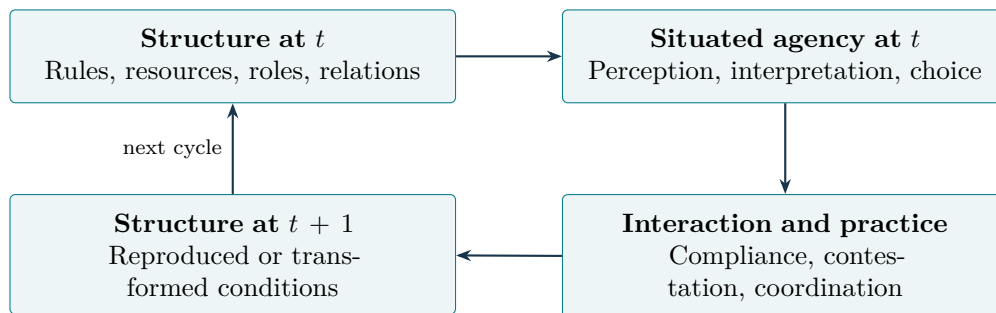
Hence:

$$\text{Actor in reality} \neq \text{Agent in model}$$

### 1.4.2 Giddens — Recursivity

Existing structure conditions action, while action reproduces or transforms subsequent structure:

$$S_t \rightarrow \text{Situated Action}_t \rightarrow S_{t+1}$$



**Figure 1. Recursive social structure.** Arrows denote conditioning and feedback, not sufficient causes. The upward arrow advances the cycle; it does not return to an unchanged past.

Persistence can give:

$$S_{t+1} \approx S_t$$

but:

$$\begin{array}{l} \text{Regularity} \not\Rightarrow \text{Linearity} \\ \text{Regularity} \not\Rightarrow \text{Determinism} \end{array}$$

Regularity, linearity, and determinism can coexist, but none should be inferred merely from observing a persistent pattern. Regularity alone establishes neither linearity nor determinism, and feedback alone does not establish chaos or unpredictability. A deterministic implementation can still generate complex trajectories. A useful **course synthesis, not a direct quotation**, is:

**History constrains the future without determining it.**

The Simon/Giddens connection is an analytical course bridge: organization helps explain persistence, while recursive practice can reproduce or transform social organization. It does not make their theories identical.

### 1.4.3 Ghorbani / Ostrom — Situate Action Institutionally

As the ontology develops, ask:

**Must institutional structure become explicit?**

Institutions structure action possibilities without mechanically determining behavior:

$$\boxed{\text{Institution} \rightarrow \text{Structured possibilities} \rightarrow \text{Agent decision} \rightarrow \text{Action}}$$

MAIA formalizes and extends IAD for conceptualizing agent-based social simulations, including explicit representation of agents, roles and institutional structures. [6] Agents need not simply comply with institutional prescriptions. [6]

## 1.5 Bottom-Up + Along the Way

The backbone remains:

$$\boxed{O_0 \longrightarrow O_1 \longrightarrow O_2 \longrightarrow O_n}$$

But along that trajectory:

|       |                   |            |                   |              |                   |             |
|-------|-------------------|------------|-------------------|--------------|-------------------|-------------|
| $O_0$ | $\longrightarrow$ | $O_1$      | $\longrightarrow$ | $O_2$        | $\longrightarrow$ | $O_n$       |
|       |                   | $\uparrow$ |                   | $\uparrow$   |                   | $\uparrow$  |
|       |                   | situate    |                   | articulate   |                   | empirically |
|       |                   | agency     |                   | institutions |                   | ground      |

The heuristic is:

**Start generatively. Construct progressively.**  
 Along the way, situate the model where the question requires it.

A richer model remains generative.

## 2 Ex Ante: Constructing the Ontology

The ex-ante question is:

**What are we going to model?**

The construction heuristic is:

**DISSECT → DISCRETIZE → COMBINE**

### 2.1 Dissect

Start from the phenomenon:

Phenomenon  $\xrightarrow{\text{dissect}}$  Relevant distinctions

**Question:**

What distinctions are necessary for investigating the phenomenon?

Dissection is analytical.

### 2.2 Discretize

Decide which distinctions become explicit modeled elements.

The first three As provide the basic vocabulary:

$A_1, A_2, A_3$

#### 2.2.1 $A_1$ — Agent

**An entity whose endogenous agency is necessary for the explanatory or decision problem.**

**Question:**

**Who has endogenous agency?**

#### 2.2.2 $A_2$ — Arena

**The material, spatial, ecological, or environmental substrate within which action occurs.**

**Question:**

**Where is action situated?**

### 2.2.3 $A_3$ — Artificiality

**Socially constructed artifacts and institutional constructs that enable, constrain, or mediate agency.**

$$A_3 = A_{3m} + A_{3s}$$

The plus sign denotes conceptual composition, not numerical addition.

#### $A_{3m}$ — Material Artificiality

Tools, technologies, infrastructure, and material artifacts that enable, constrain, or mediate action. A ball or a video-review device can be represented here.

#### $A_{3s}$ — Social Artificiality

Institutions, norms, formal rules, roles, and authority arrangements. Rules of play, officiating roles, and review permissions can be represented here.

These distinctions classify modeled aspects. VAR equipment belongs to  $A_{3m}$ ; the protocol specifying when review is permitted and who can overturn a decision belongs to  $A_{3s}$ . Their connections to agents and other elements belong to  $A_4$ . A technology can therefore involve both material and institutional representations.

#### Behavioral mechanisms and social prescriptions

Every executable ABM requires behavioral and interaction mechanisms. A movement procedure is an operationalization of agent capabilities; it is not automatically a social institution. A prohibition on an action, the authority to sanction it, and agents' responses to that prescription introduce  $A_{3s}$ . Classify the represented content, not whether it happens to be implemented as code.

A useful heuristic is:

$$\text{Artificiality} = \text{affordance and/or barrier}$$

## 2.3 Combine: $A_4$ — Array

Discrete elements must be arranged into a system.

**Array is the assumed systemic arrangement organizing Agents, Arena and Artificiality.**

$$A_4 = C(A_1, A_2, A_3)$$

It can include:

- relations,
- networks,
- topology,
- hierarchy,
- spatial configurations,
- dependencies,
- institutional configurations.

Thus:

$$\underbrace{A_1 + A_2 + A_3}_{\text{DISCRETIZATION}} + \underbrace{A_4}_{\text{COMBINATORICS}} = \text{ONTOLOGY}$$

The  $A$ s classify **modeling roles**, not intrinsic properties of real-world entities.

## 2.4 LUL: Making the Ontology Explicit

**Light-UML-Like (LUL)** represents:

WHAT IS?

Its core vocabulary is:

Classes + State + Capabilities + Relations

using stereotypes such as:

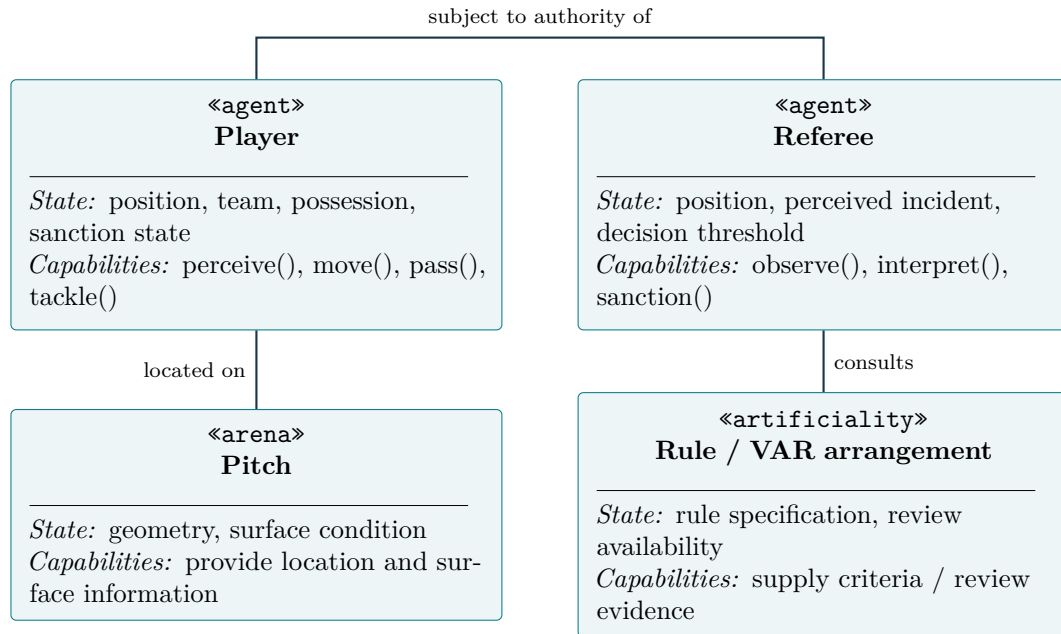
«agent» «arena» «artificiality»

LUL represents the **modeled world**, not the simulation software.

Therefore it excludes:

- multiplicity,
- CLOCK,
- scheduling,
- setup / go,
- simulation controllers,
- implementation architecture,
- emergent outcomes.

Represent the modeled world, not the simulation software.



**Figure 2. A LUL fragment with explicit Artificiality and referee agency.** Grouping Rule / VAR saves space; they can be separate classes. This fragment represents explicit enforcement and referee discretion. Rules and review authority are  $A_{3s}$ ; VAR equipment is  $A_{3m}$ . The question determines whether this detail is needed. Relations have no temporal direction. Non-agent capabilities describe available functions or affordances, not autonomous decisions.

### 3 Progressive Ontologies: From I-ABM to P-ABM

#### 3.1 Baseline Ontology $O_0$

$O_0$  is the deliberately minimal reference ontology from which progressive construction begins.

Start here for comparison  $\neq$  Reality fundamentally starts here

#### 3.2 I-ABM — Intuitive ABM

An I-ABM is a deliberately parsimonious generative model centered on agents, their arena, and their arrangement. It avoids unnecessary technological representation and retains only the social rules needed to investigate the phenomenon.

$$\underbrace{A_1 + A_2 + A_4}_{\text{generative core}} + \underbrace{\text{question-required } A_3}_{\text{selective representation}}$$

This expression summarizes ontological commitments, not a complete executable model: dynamics and initialization must also be specified.

##### Two principles of parsimony

**Material Artificiality:** avoid explicitly representing technology and material artifacts when they are unnecessary to the generative question. Include them when their mediation matters.

**Social Artificiality:** retain a minimal sufficient set of social rules and institutional arrangements. “Minimal” is relative to the question, not a fixed number of rules.

Neither component is required to be absent. These are modeling preferences rather than numerical limits or an optimization algorithm. A ball may be indispensable to a minimal passing model; an ownership norm may be indispensable to a minimal institutional model. The inclusion of either does not by itself make the model a P-ABM.

**What can these agents generate with the minimum material and social conditions needed to investigate the phenomenon?**

##### 3.2.1 Course interpretation of the readings

The following synthesis connects the Four As to the readings after the categories have been introduced. It is a construction heuristic developed for this course, not terminology or a taxonomy proposed by the authors.

| Contribution                      | Construction emphasis                                                                      | Question for the modeler                                                                                 |
|-----------------------------------|--------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| Epstein: generate                 | Center $A_1 + A_2 + A_4$ as the generative core.                                           | What can local agent interactions generate?                                                              |
| Cioffi: construct progressively   | Re-specify all four As; retain only question-required A3.                                  | What must be added, removed, differentiated, reclassified, or recombined?                                |
| Ghorbani: articulate institutions | Elaborate $A_{3s}$ as needed, keeping $A_{3m}$ parsimonious where possible.                | Which roles, prescriptions, permissions, and enforcement arrangements must be explicit?                  |
| Knoeri: situate empirically       | Ground agency and relevant conditions across all four As; seek contextually sufficient A3. | Which actors, decisions, environments, artifacts, institutions, and relations require empirical support? |

Epstein's generative approach does not exclude institutions; Ghorbani's MAIA does not prescribe minimizing technology. Those emphases express our teaching strategy [3, 6]. Cioffi provides a revision process across the ontology, not a prescribed amount of A3 [4]. Knoeri strengthens the contextual and empirical specification of agents and their decisions as well as their surroundings [5].

#### Increasingly explicit and justified situatedness

The progression seeks a contextually sufficient representation, not the fullest possible inventory. A richer ontology remains selective and generative. Institutional articulation and empirical grounding can occur together; they are not compulsory consecutive stages.

### 3.3 Progressive Ontological Re-Specification

As the question or purpose changes:

$$O_i \xrightarrow{\text{re-specification}} O_{i+1}$$

using:

$$\text{ADD} + \text{REMOVE} + \text{DIFFERENTIATE} + \text{RECLASSIFY} + \text{RECOMBINE}$$

An entity can therefore change modeling role:

$$A_i \rightarrow A_j$$

This is **ontological migration within the model**, not a claim about the entity's intrinsic nature.

### 3.4 Along the Way: Ontology, Purpose, and Policy Use

#### Three distinct judgments

**Ontology:** What entities, relations, and mechanisms are explicitly represented?

**Purpose:** Is the model exploring mechanisms or informing a particular intervention decision?

**Evidential warrant:** What claims do its assumptions, experiments, and assessment support? A change in ontology does not automatically change purpose; a policy purpose does not automatically establish evidential warrant.

### 3.4.1 Exploratory models and policy purpose

An I-ABM can explicitly represent material or social Artificiality when the generative question requires it. Further exploration may enrich that representation without changing the model's purpose. There is no categorical boundary at the first introduction of A3.

| Model description               | Representational preference                                           | Purpose and warrant                                                                            |
|---------------------------------|-----------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| I-ABM                           | Center A1, A2, and A4; use minimal sufficient $A_{3m}$ and $A_{3s}$ . | Explore generative mechanisms with deliberately restricted commitments.                        |
| Further exploratory development | Re-specify relevant elements across all four As.                      | Deepen mechanism inquiry; richer representation alone confers no policy warrant.               |
| P-ABM                           | Justify representation for the intervention question.                 | Support conditional prospective claims, subject to evidence and decision-context requirements. |

### 3.4.2 Moving toward policy use

Three questions become progressively important:

1. Is agency adequately situated and empirically specified?
2. Which material and institutional conditions require explicit representation?
3. Are the assumptions and experiments defensible for the intended claim?

Generative exploration  $\xrightarrow{\text{policy purpose and additional evidential requirements}}$  P-ABM.

A project may remain exploratory or begin from a policy question. Cioffi-inspired re-specification can revise agents, arena, Artificiality, and their arrangement throughout either path. The transition toward policy use does not eliminate generativity; it increases the need to justify the model's situatedness and the claims drawn from it.

## 3.5 P-ABM — Policy ABM

The P-ABM asks a stronger question:

**What can situated agents generate under empirically defensible institutional, structural and intervention conditions?**

Its purpose is prospective:

Alternative actions/conditions  $\rightarrow$  Alternative possible futures

This stronger purpose will eventually impose a stronger **ex-post evidential burden**.

### 3.6 Soccer Example

#### 3.6.1 I-ABM: a minimal passing model

$$\begin{aligned}
 A_1 &= \{\text{Players}\}, \\
 A_2 &= \{\text{Field}\}, \\
 A_{3m} &= \{\text{Ball}\}, \\
 A_{3s} &= \{\text{minimal rules of play relevant to the question}\}, \\
 A_4 &= \{\text{spatial, team-membership, and interaction arrangements}\}.
 \end{aligned}$$

The ball is required for this passing question; goals need appear only if scoring matters. VAR and other unnecessary technologies can be omitted. The minimum social rules might distinguish permitted passes between teammates from prohibited handling; retain only the prescriptions whose behavioral consequences the experiment investigates. Specify separately how agents choose movements and passes, and whether they can violate represented prescriptions.

#### Soccer laboratory: parsimony without categorical exclusion

A movement-only question might need no explicit ball. A passing question does. Both can motivate an I-ABM because the criterion is selective representation for generative inquiry, not the absence of Artificiality. Team membership belongs to  $A_4$ ; the prescriptions attached to a team or officiating role belong to  $A_{3s}$ .

#### 3.6.2 Along the way: institutions and technology

If enforcement becomes relevant, make the required laws, sanctions, and officiating roles explicit in  $A_{3s}$ . If video review becomes relevant, distinguish its equipment in  $A_{3m}$  from its review protocol and decision authority in  $A_{3s}$ . Elaborate either only to the extent needed by the question.

#### 3.6.3 Endogenizing discretion and grounding agency

A simpler version may represent the referee as an exogenous enforcement service in  $A_{3s}$ . When perception, interpretation, or discretion matters, replace that service with an  $A_1$  referee and specify its information, state, capabilities, and decisions.

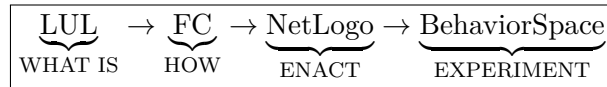
$$\text{Referee-as-service}_{A_{3s}} \xrightarrow{\text{endogenize discretion}} \text{Referee-as-agent}_{A_1}.$$

Retain the officiating role and laws in  $A_{3s}$ , and authority and information relations in  $A_4$ . Do not preserve a duplicate automatic sanction mechanism. Contextual grounding may require evidence about players' choices, referee perception, field conditions, and relevant relations, not merely more detailed artifacts.

The ontology has been re-specified, not simply enlarged. A policy study of an enforcement intervention would additionally need defensible evidence and an explicit decision context. Introducing a ball, rules, or referee agency alone does not establish policy validity.

## 4 From Ontology to Simulation

The operational progression is:



### 4.1 LUL $\rightarrow$ Flowchart

Capabilities become processes:

$$f() \rightarrow [f]$$

States become conditions and updates:

$$\text{State} \rightarrow \{\text{Condition}, \text{Update}\}$$

Relations become operational interactions.

Thus:

|                                    |
|------------------------------------|
| LUL: WHAT IS $\rightarrow$ FC: HOW |
|------------------------------------|

### 4.2 CLOCK

Flowcharting introduces temporal organization:

|              |
|--------------|
| <b>CLOCK</b> |
|--------------|

including:

- iteration,
- activation,
- scheduling,
- serialization,
- parallelization,
- synchronization,
- termination.

Therefore:

|                                        |
|----------------------------------------|
| $A_4 = \text{HOW THINGS ARE ARRANGED}$ |
|----------------------------------------|

whereas:

|                                 |
|---------------------------------|
| $FC = \text{HOW THINGS HAPPEN}$ |
|---------------------------------|

So:

|                                          |
|------------------------------------------|
| $FC \neq \text{LUL redrawn with arrows}$ |
|------------------------------------------|

### 4.3 Implementation

$$\boxed{\text{LUL} \rightarrow \text{FC} \rightarrow \text{NetLogo}}$$

If implementation exposes conceptual ambiguity, move backward:

$$\boxed{\text{NetLogo} \rightarrow \text{FC} \rightarrow \text{LUL}}$$

### 4.4 Emergence

For a run:

$$\boxed{M + I_0 + \theta + \omega \rightarrow T}$$

where  $M$  is the model,  $I_0$  the initial conditions,  $\theta$  its parameterization,  $\omega$  a stochastic realization, and  $T$  the resulting trajectory.

The modeller specifies the left-hand side.

The simulation generates  $T$ .

$$\boxed{\text{ABM specification ends before the emergent outcome begins.}}$$

### 4.5 From Runs to Behavioral Space

One run is not the behavior of the model:

$$\boxed{\text{One run} \neq \text{the model's behavior}}$$

Instead:

$$\boxed{\mathcal{B}(M) = \{T(M, I_0, \theta, \omega)\}}$$

We therefore move:

$$\boxed{\text{Run} \rightarrow \text{Replications} \rightarrow \text{Parameter experiments} \rightarrow \text{Behavioral space}}$$

BehaviorSpace is machinery for generating systematic simulation evidence. It is **not itself the analysis of that evidence**.

## 5 Ex Post: Documenting for Auditability and Actionability

Ex post, the question changes:

**What did we build, what did we do with it,  
and what can we defensibly claim from it?**

There are two levels:

ALL ABMs → **AUDITABILITY**

and:

P-ABMs → **AUDITABILITY + ACTIONABILITY**

### 5.1 Documenting for Auditability — All ABMs

Auditability is the minimum ex-post requirement.

**AUDITABILITY** = Transparency + Traceability + Replicability + Reproducibility

#### 5.1.1 Transparency

Are the model's assumptions, entities, processes, parameters, initialization, scheduling and outputs explicit?

#### 5.1.2 Traceability

Can we follow:

Ontology → Implementation → Experiment → Result → Claim

#### 5.1.3 Replicability

Can the specified computational experiment be rerun and its reported behavior recovered within the appropriate stochastic tolerance?

#### 5.1.4 Reproducibility

Are the model, data, assumptions, procedures and analytical operations sufficiently explicit and available for independent computational scrutiny?

These are the course's operational definitions. Because *replicability* and *reproducibility* are used differently across disciplines, these operational definitions should be made explicit in the course.

### 5.2 Documenting for Actionability — P-ABMs

A P-ABM carries an additional burden because it is intended to support reasoning about possible interventions and futures.

$P\text{-ABM} \Rightarrow \text{Auditability} + \text{Prospective evidential burden}$

A selected run cannot satisfy that burden.

Neither can a convenient scenario.

The model's evidence must be systematically interrogated.

### 5.3 Meta-Analysis of Simulation Evidence

We use **meta-analysis of simulation evidence** for the systematic ex-post assessment of the model and the body of computational evidence it generates.

#### Terminology: meta-analysis of simulation evidence

This is a course term for systematic assessment across a model and its computational experiments. It is broader than conventional statistical meta-analysis across independent empirical studies: it need not pool published effect sizes or treat simulation runs as independent studies. Whenever this guide abbreviates the phrase to “meta-analysis,” this broader meaning applies.

The constituent methods should remain explicit. Replications characterize stochastic variation; parameter sweeps organize exploration; sensitivity and robustness analyses assess dependence on assumptions; scenario comparisons support conditional claims. Calling the collection meta-analysis does not replace those analytical tasks. These methods can also strengthen exploratory ABMs; P-ABMs carry an additional obligation to connect their results to prospective claims.

The analytical object is no longer an individual simulation result:

Runs  $\rightarrow$  Experiments  $\rightarrow$  Simulation evidence  $\rightarrow$  **META-ANALYSIS**

It comprises:

#### 5.3.1 Verification

Did we implement the model as specified?

#### 5.3.2 Validation

Is the model adequate for its stated purpose and domain?

#### 5.3.3 Replications

Characterize stochastic variation:

$$M + I_0 + \theta + \omega_r \rightarrow T_r$$

#### 5.3.4 Parameter Sweeps

Explore relevant parameter space:

$$\theta_1, \theta_2, \dots, \theta_n$$

with replications within parameterizations:

$$M + I_0 + \theta_j + \omega_r \rightarrow T_{jr}$$

### 5.3.5 Sensitivity Analysis

Determine which assumptions, parameters and initial conditions materially influence outcomes.

### 5.3.6 Robustness Analysis

Determine which findings persist across plausible specifications and uncertainties.

### 5.3.7 Scenario Analysis

Define theoretically or policy-relevant configurations:

$$S_j = (P_j, \theta_j, I_j)$$

Here  $P_j$  denotes the intervention or policy configuration,  $\theta_j$  the parameters, and  $I_j$  the initial conditions. Hold the model specification explicit when comparing scenarios, and characterize their conditional behavioral spaces:

$$\mathcal{B}(S_j) = \{T_{j1}, T_{j2}, \dots, T_{jR}\}$$

### 5.3.8 Cross-Scenario Analysis

Compare:

$$\mathcal{B}(S_1), \mathcal{B}(S_2), \dots, \mathcal{B}(S_n)$$

to determine how alternative interventions and conditions alter possible outcomes.

## 5.4 Prospective Claims

The objective of a P-ABM need not be point prediction. The distinction below separates a point forecast from the broader conditional claims used in this course; it does not deny that predictions can also be conditional.

Distinguish:

$$\boxed{\text{A prospective claim need not be a point prediction}}$$

A prospective claim is conditional:

$$\boxed{\text{Conditions} + \text{Intervention} \rightarrow \text{Structured set of possible futures}}$$

Instead of:

$$P \text{ will produce } Y,$$

the defensible form may be:

**Under specified assumptions and conditions,  $P$  changes the distribution or structure of possible outcomes in the following ways.**

## 5.5 Actionability

Prospective claims can support action when their evidential warrant is adequate and they address a specified decision context. Surviving scrutiny of simulation evidence is necessary, but does not by itself establish what should be done.

Simulation evidence  $\xrightarrow{\text{systematic assessment}}$  Warranted prospective claims  
 Warranted claims + Decision context  $\longrightarrow$  Potential actionability.

### 5.5.1 Specify the decision context

- **Decision-maker and authority:** Who can act, and what lies within their control?
- **Alternatives and feasibility:** Which interventions are available, compared with what baseline, under what resource and implementation constraints?
- **Objectives and trade-offs:** Which outcomes matter, to whom, and how are competing objectives and distributional consequences evaluated?
- **Domain and uncertainty:** Where and over what horizon is the claim applicable? Which uncertainties, risks, and failure conditions remain material?

These judgments should be documented alongside the evidence. Simulation can inform trade-offs; it does not determine whose objectives should prevail or who has legitimate authority to decide.

#### Soccer laboratory: from a finding to a decision

A model might investigate whether a review procedure changes the distribution of incorrect calls, interruptions, or unequal effects across teams. To make a resulting conditional claim useful for a competition organizer, identify feasible review procedures, staffing and technology constraints, and the trade-off between accuracy and interruption. No policy effect is asserted here: this is the information a decision-oriented study would need to establish and assess.

### 5.5.2 Examine the evidence and its limits

Meta-analysis should therefore examine:

- robustness,
- sensitivity,
- stochastic variation,
- initial-condition sensitivity,
- thresholds,
- regime changes,
- path dependence,
- extreme outcomes,
- unintended consequences,
- distributional consequences,
- conditions of success and failure.

Thus the policy question is not simply:

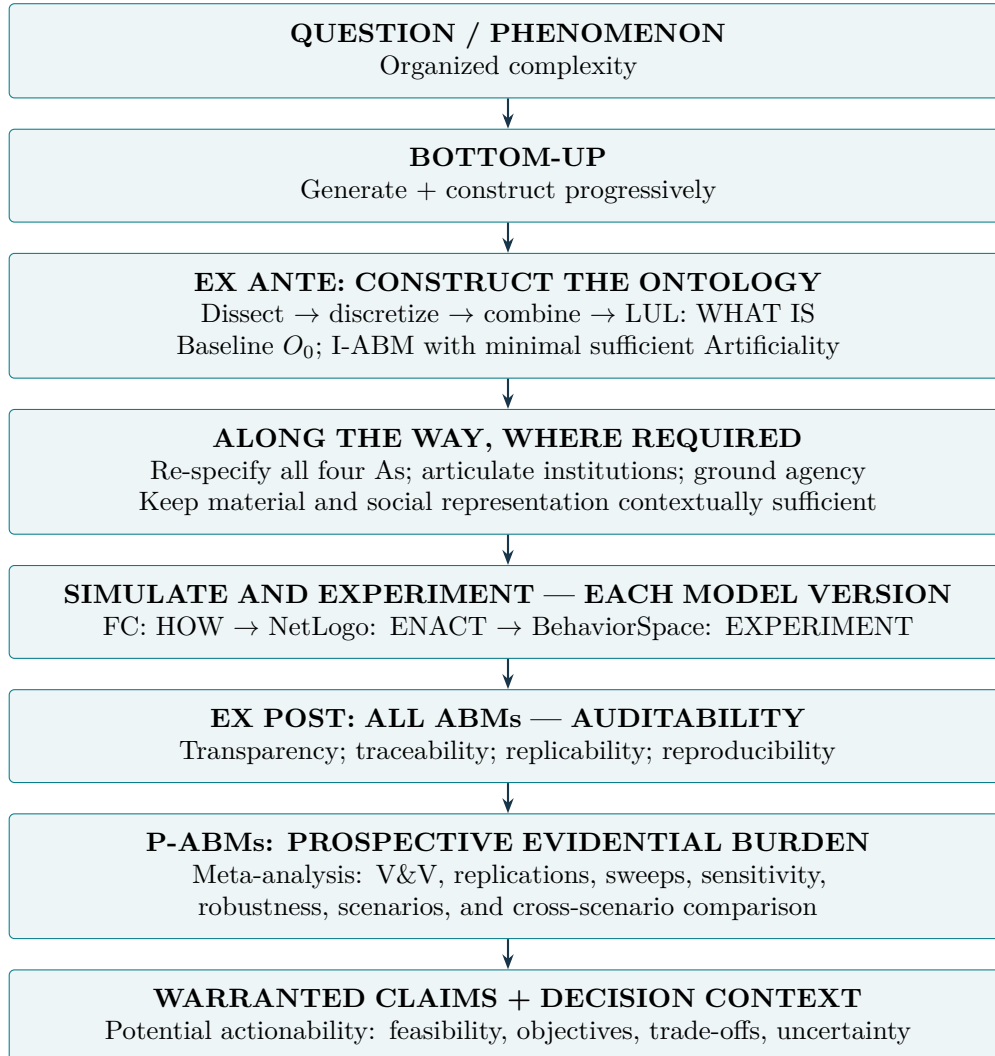
Does intervention  $P$  work?

It becomes:

**Under what conditions does  $P$  alter possible futures,  
how robustly, for whom, and with what vulnerabilities?**

## 6 The ABM Construction Method

### 6.1 Complete Architecture



**Figure 3. Complete methodological architecture.** The model can be executed and assessed at each revision. P-ABM is a purpose-dependent extension; it is not a prerequisite for simulation. Assessment can lead back to construction.

### 6.2 Construction Protocol

The guide can finally be reduced to a working sequence:

#### THINK

What organized complexity matters?

#### DISSECT

What distinctions matter?

#### DISCRETIZE

What becomes  $A_1, A_2, A_3$ ?

#### COMBINE

How are they arranged through  $A_4$ ?

**SPECIFY**

What is the ex-ante ontology?

**GENERATE**

What can agents generate with minimal sufficient material and social conditions?

**RE-SPECIFY**

What needs to change along the way?

**SITUATE**

Which agents, decisions, material artifacts, institutions, and relations require explicit representation and empirical grounding? Distinguish this revision from a change to policy purpose.

**OPERATIONALIZE**

How does the system work through time?

**ENACT**

Implement the computational model.

**EXPERIMENT**

Generate systematic simulation evidence.

**AUDIT**

Can the model and evidence be scrutinized?

**META-ANALYZE**

For P-ABM, what does the body of evidence support?

**CLAIM**

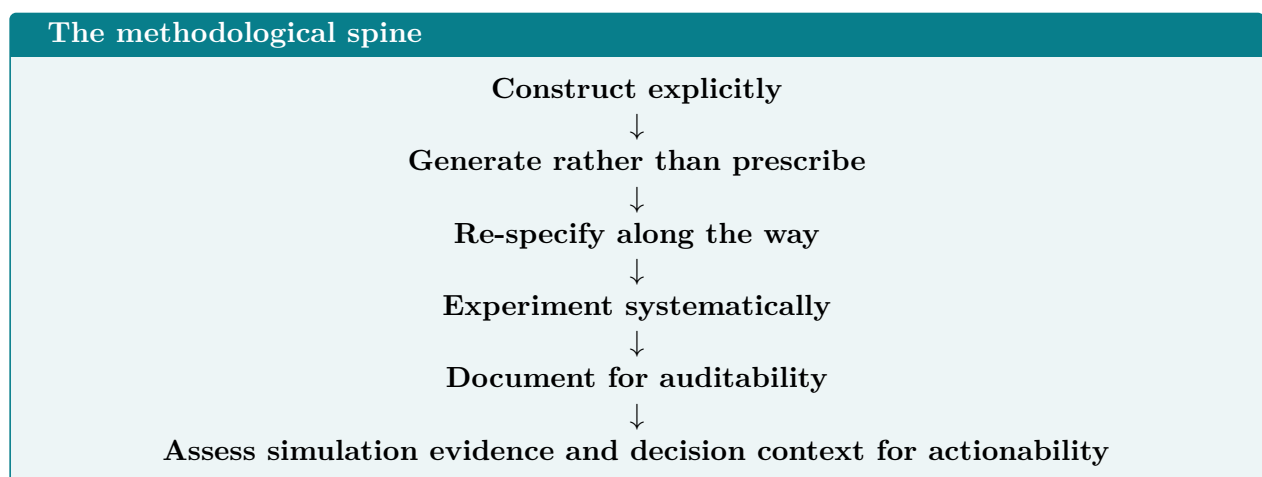
What prospective claims are defensible?

**ACT**

For whom is the claim actionable, among which feasible alternatives, under what objectives, trade-offs, and uncertainties?

Generation is a recurring goal: executable generative experiments require the operationalization and enactment steps below it. V&V informs revision but does not provide an algorithm for constructing  $M_{i+1}$ .

The methodological spine is therefore:



## References and source boundaries

The six entries below are the core readings used in Section 1. Giddens and Ostrom are discussed through Knoeri et al. and Ghorbani et al.; Lakatos is discussed through Cioffi-Revilla. This document does not claim independent consultation of their original books.

## References

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- [2] Simon, H. A. (1962). The architecture of complexity. *Proceedings of the American Philosophical Society*, 106(6), 467–482. Uploaded reading: [Simon\\_1962.pdf](#).
- [3] Epstein, J. M. (1999). Agent-based computational models and generative social science. *Complexity*, 4(5), 41–60. Uploaded reading: [epstein-complexity-1999.pdf](#).
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- [6] Ghorbani, A., Bots, P., Dignum, V., & Dijkema, G. (2013). MAIA: A framework for developing agent-based social simulations. *Journal of Artificial Societies and Social Simulation*, 16(2), 9. Uploaded reading: [amineh\\_etal\\_2013.pdf](#). <https://jasss.soc.surrey.ac.uk/16/2/9.html>

### How to read the attributions

**Reading-grounded:** organized complexity; hierarchical organization and near-decomposability; generative explanation and its limits; progressive model sequences; context-specific situated agency; institutional action and possible non-compliance.

**Course synthesis:** bottom-up + along the way; the history statement and Simon/Giddens bridge; Four As; the parsimonious I-ABM heuristic and the distinction between  $A_{3m}$  and  $A_{3s}$ ; progressive re-specification; restricted LUL; auditability and actionability; broad meta-analysis of simulation evidence; prospective claims; and the soccer illustrations.